Module 1 solution

```
source("R/prepare_everything.R")
```

```
[1] "Metadata imported!"
[1] "Chronos scores imported!"
```

```
[1] "Expression data imported!"
[1] "Mutation data imported!"
```

Modul 1: Sex chromosome biology: Do models annotated as male or female differ in expression of Y-linked genes?

`chr_y_genes` contains genes located on the Y chromosome:

```
chr_y_genes
```

```
[1] "DDX3Y"  "EIF1AY" "KDM5D"  "RPS4Y1" "UTY"    "ZFY"
```

1.1 Explore the patient data and find out where to find out information on the sex/gender. How many samples do we have for each sex? Do we have this info for all patients?

```
models[, .N, by = Sex]
```

```
       Sex     N
    <char> <int>
1:  Female  1025
2:    Male  1351
3: Unknown   127
4:            25
```

1.2 Compute the expression of all `chr_y_genes`, independently for each sex. We don't need to look at every gene individually. What do you observe? Find a nice way to visualize that.

```
female <- models[ModelID %in% rownames(expression) & Sex == "Female", ModelID]
male <- models[ModelID %in% rownames(expression) & Sex == "Male", ModelID]
unknown <- models[ModelID %in% rownames(expression) & Sex %in% c("", "Unknown"), ModelID]

print(paste0("Female: ",  mean(expression[female,chr_y_genes], na.rm = TRUE)))
```

```
[1] "Female: 0.101885076736916"
```

```
print(paste0("Male: ", mean(expression[male,chr_y_genes], na.rm = TRUE)))
```

```
[1] "Male: 2.78816172183949"
```

```
print(paste0("Unknown: ", mean(expression[unknown,chr_y_genes], na.rm = TRUE)))
```

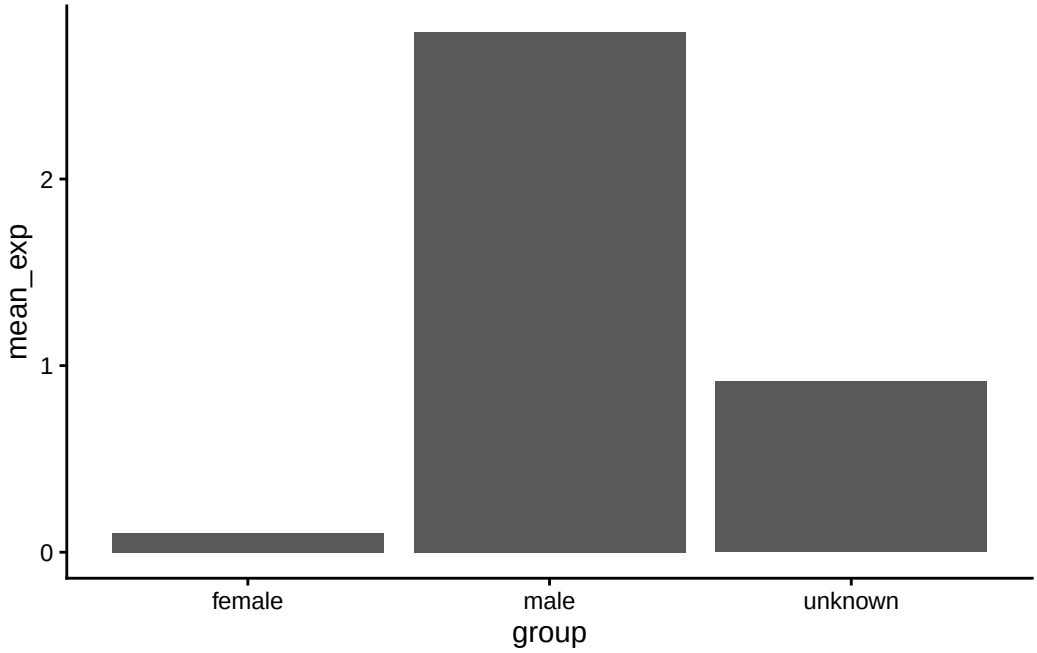
```
[1] "Unknown: 0.916280142318437"
```

```
plot_data <- data.table(
  group = c("female", "male", "unknown"),
  mean_exp = c(mean(expression[female,chr_y_genes]),
               mean(expression[male,chr_y_genes]),
               mean(expression[unknown,chr_y_genes])))

plot_data
```

```
     group  mean_exp
    <char>     <num>
1:  female 0.1018851
2:    male 2.7881617
3: unknown 0.9162801
```

```
ggplot(plot_data, aes(group, mean_exp)) +
  geom_col()
```

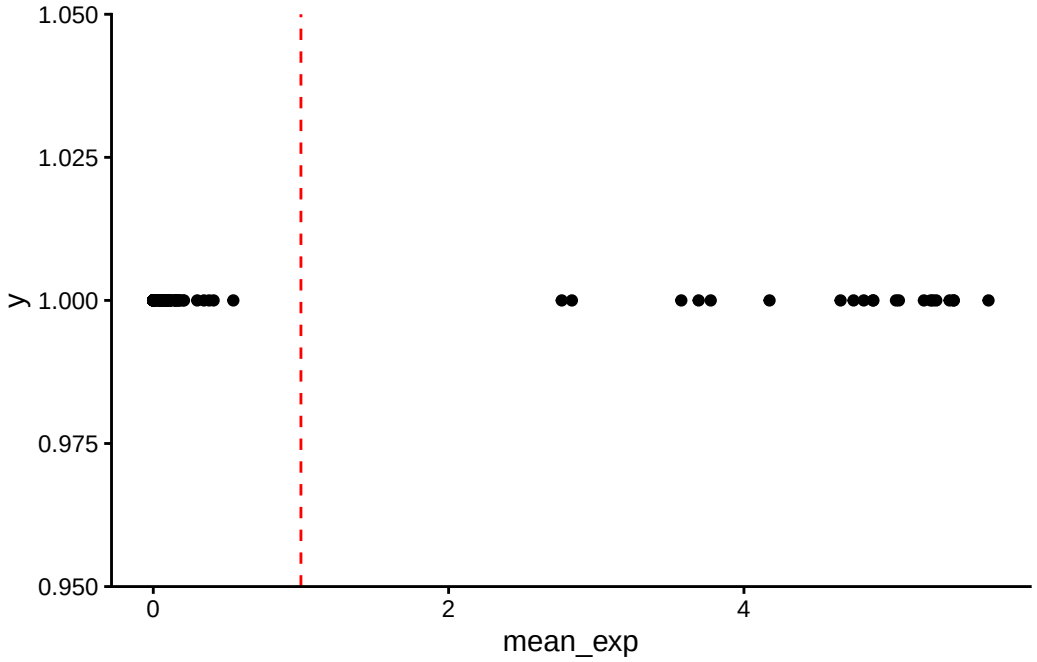


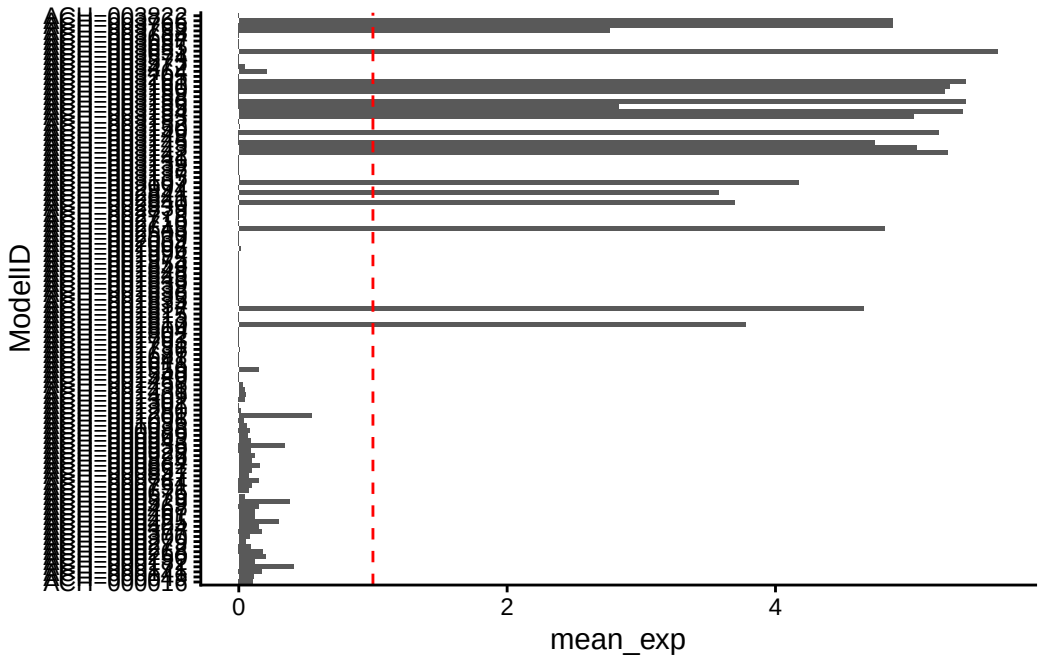
1.3 Lets assume we explore a research question that requires as many female patients as possible. We can use the 1025 with assigned metadata, but we want to look for female patients among the patients with unknown or missing sex information. Can we find out (or at least make an educated guess) if a patient is likely male or female? Find a way to visualize that.

```
plot_data <- models[Sex %in% c("Unknown", "") & ModelID %in% rownames(expression)]

plot_data[, mean_exp := apply(expression[ModelID,chr_y_genes], 1, mean, na.rm = TRUE)]

ggplot(plot_data,
  aes(mean_exp, 1)) +
  geom_point() +
  geom_vline(xintercept = 1, linetype = "dashed", color = "red")
```




```
ggplot(plot_data, aes(mean_exp, ModelID)) +
  geom_col() +
  geom_vline(xintercept = 1, linetype = "dashed", color = "red")
```